

1.PALINDROME

SIMATS Online Programming Lab
JAVA PROGRAMMING

BANDLAPALLI BHANUTEJA REDDY
192325016RD

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Q1 Palindrome number/string 10 marks

Palindrome number/string

Your Code

```
1 import java.util.Scanner;
2 class Main{
3     public static void main(String[]args){
4         Scanner sc=new Scanner(System.in);
5         String str=sc.next();
6         String rev="";
7         for(int i=str.length()-1;i>=0;i--){
8             rev=rev+str.charAt(i);
9         }
10        if(str.compareTo(rev)==0)
11            System.out.println("Palindrome");
12        else
13            System.out.println("Not Palindrome");
14    }
15 }
```

Input
TET

Output
Palindrome

2.FIRST N PERFECT NUMBERS

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Q2 First n perfect numbers 10 marks

First n perfect numbers

Your Code

```
1 import java.util.Scanner;
2 class Main{
3     public static void main(String[]args){
4         Scanner sc=new Scanner(System.in);
5         int n=sc.nextInt();
6         int count=0;
7         int num=1;
8         while(count<n){
9             int sum=0;
10            for(int i=1;i<=num/2;i++){
11                if(num%i==0){
12                    sum+=i;
13                }
14            }
15            if(sum==num){
16                System.out.print(num+" ");
17                count++;
18            }
19            num++;
20        }
21    }
22 }
```

Input
4

Output
6 28 496 8128

Run Save

3.DECIMAL TO OCTAL AND BINARY

Q3 Decimal number equivalent to binary number and octal number10 marks

Decimal number equivalent to binary number and octal number

Your Code

```
1 import java.util.Scanner;
2 class Main{
3 public static void main(String[] args){
4 Scanner sc=new Scanner(System.in);
5 int bin=sc.nextInt();
6 int decimal1=0;
7 int base=1;
8 while(bin>0){
9 int rem=bin%10;
10 decimal1=decimal1+(rem*base);
11 base=base*2;
12 bin=bin/10;
13 }
14 int oct=sc.nextInt();
15 int decimal2=0;
16 base=1;
17 while(oct>0){
18 int rem=oct%10;
19 decimal2=decimal2+(rem*base);
20 base=base*8;
21 oct=oct/10;
22 }
23 System.out.println("Binary to Decimal = "+decimal1);
24 System.out.println("Octal to Decimal = "+decimal2);
25 }
26 }
```

Input
1011
17

Output
Binary to Decimal = 11
Octal to Decimal = 15

4.NTH MAX AND NTH MIN

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Q4 Nth max number and nth min number in array then sum of it and difference of it10 marks

Nth max number and nth min number in array then sum of it and difference of it

Your Code

```
1 import java.util.Scanner;
2 class Main {
3 public static void main(String[] args) {
4 Scanner sc=new Scanner(System.in);
5 int n=sc.nextInt();
6 int[] arr=new int[n];
7 for (int i=0;i<n;i++) {
8 arr[i]=sc.nextInt();
9 }
10 int k=sc.nextInt();
11 for (int i=0;i<n-1;i++) {
12 for (int j=i+1;j<n;j++) {
13 if (arr[i]>arr[j]) {
14 int temp=arr[i];
15 arr[i]=arr[j];
16 arr[j]=temp;
17 }
18 }
19 }
20 int nthMin=arr[k-1];
21 int nthMax=arr[n-k];
22 int sum=nthMin+nthMax;
23 int difference=nthMax-nthMin;
24 System.out.println("Nth Minimum: "+nthMin);
25 System.out.println("Nth Maximum: "+nthMax);
26 System.out.println("Sum: "+sum);
27 System.out.println("Difference: "+difference);
28 }
29 }
```

Input
7
10 5 20 30 15 25 40
2

Output
Nth Minimum: 10
Nth Maximum: 30
Sum: 40
Difference: 20

5. AVERAGE OF POSITIVE AND NEGATIVE

Q5 Find the average positive and negative numbers upto -110 marks

Find the average positive and negative numbers upto -1

Your Code

```
1 import java.util.Scanner;
2 class Main {
3     public static void main(String[] args) {
4         Scanner sc = new Scanner(System.in);
5         int posSum = 0, negSum = 0;
6         int posCount = 0, negCount = 0;
7         int num = sc.nextInt();
8         while (num!=-1) {
9             if (num>0) {
10                 posSum+=num;
11                 posCount++;
12             } else {
13                 negSum+=num;
14                 negCount++;
15             }
16             num=sc.nextInt();
17         }
18         if(posCount>0) {
19             System.out.println("Average of positive numbers: "+(double)
20         } else{
21             System.out.println("No positive numbers");
22         }
23         if(negCount>0){
24             System.out.println("Average of negative numbers: "+(double)
25         } else {
26             System.out.println("No negative numbers");
27         }
```

Input

```
10
20
-5
15
..
```

Output

```
Average of positive numbers: 17.5
Average of negative numbers: -7.5
```

6.REMOVING DUPLICATES IN ARRAY

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Q6 Removing duplicates in array10 marks

Removing duplicates in array

Your Code

```
1 import java.util.Scanner;
2 class Main{
3     public static void main(String[] args){
4         Scanner sc = new Scanner(System.in);
5         int n = sc.nextInt();
6         int arr[] = new int[n];
7
8         for(int i=0;i<n;i++){
9             arr[i] = sc.nextInt();
10        }
11        for(int i=0;i<n;i++){
12            for(int j=i+1;j<n;j++){
13                if(arr[i]==arr[j]){
14                    for(int k=j;k<n-1;k++){
15                        arr[k]=arr[k+1];
16                    }
17                    n--;
18                    j--;
19                }
20            }
21        }
22        System.out.println("Array without duplicates: ");
23        for(int i=0;i<n;i++){
24            System.out.print(arr[i]+" ");
25        }
26    }
27 }
```

Input

```
6
4 5 6 5 7 8
```

Output

```
Array without duplicates:
4 5 6 7 8
```

7.MATRIX MULTIPLICATION

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Q7 Matrix multiplication 10 marks

Matrix multiplication

Your Code

```
1 import java.util.Scanner;
2 class Main {
3     public static void main(String[] args) {
4         Scanner sc = new Scanner(System.in);
5         int r = sc.nextInt();
6         int c = sc.nextInt();
7
8         int[][] A = new int[r][c];
9         int[][] B = new int[r][c];
10        int[][] C = new int[r][c];
11
12        for(int i=0;i<r;i++){
13            for(int j=0;j<c;j++){
14                A[i][j]=sc.nextInt();
15            }
16        }
17
18        for(int i=0;i<r;i++){
19            for(int j=0;j<c;j++){
20                B[i][j]=sc.nextInt();
21            }
22        }
23
24        for(int i=0;i<r;i++){
25            for(int j=0;j<c;j++){
26                C[i][j]=0;
27                for(int k=0;k<c;k++){
28                    C[i][j] += A[i][k]*B[k][j];
29                }
30            }
31        }
32    }
33 }
```

Input

```
3 3
1 2 3
4 5 6
7 8 9
```

Output

```
Result matrix:
18 24 30
54 69 84
90 114 138
```